

Sprint 4 Report

Product Name: SlugRoute

Date: June 3, 2026

Revision: 1

Actions to stop doing

- **Stop hardcoding configuration dates:** While building our recurrence and date checking logic, we occasionally relied on hardcoded boundary values. We must stop manual date-configuration overrides and shift toward fully automated date-boundary calculations from the database.
- **Stop over-complicating waypoint paths:** In our initial multi-stop routing builds, we attempted to draw separate independent polylines, which resulted in map rendering lag. We stopped using multiple map update loops and moved to a single centralized polyline renderer.
- **Stop skipping integration test updates:** During our rush to implement the calendar export service and the Go API proxy, we deferred updating backend unit tests until the end of the sprint. We should stop writing tests as an afterthought and write them in parallel with API changes.

Actions to start doing

- **Adopt a Cross-Device and Cross-Browser Testing Protocol:** To prevent mobile layout regressions and platform-specific download errors, the team should adopt a standard multi-device testing matrix (iOS, Android, and desktop browsers) as a mandatory checkpoint before any user-facing feature is considered "Done."
- **Establish Database Query Profiling and Performance Safeguards:** As our data set grows, we must start proactively profiling database queries during feature design to prevent latency bottlenecks under high concurrent loads, rather than waiting for query delays to occur in production.
- **Enforce Automated Code Linting and Formatting Checkpoints:** To maintain code consistency and minimize cognitive overhead during peer reviews, we should adopt automated formatting pipelines (using pre-commit hooks and static analysis tools) across Go and Python to enforce our style guides programmatically.

Actions to keep doing

- **Keep using specific Task IDs in commit messages:** This practice remained essential for maintaining traceability on our Scrum board as we integrated multiple security, calendar, and routing tasks, helping the team review chronological changes.

- **Keep the "Touch-Target First" design standard:** Following our mobile spacing and CSS audit, maintaining generous interactive targets ensured the continuous routing buttons and map overlays remain legible and easy to use on small screens.
- **Keep the Sunday "Integration Sync":** The 12pm–2pm Sunday meeting remains our most effective window for bridging the interface boundary between our Go calendar service and the JavaScript client.

Work completed

- **User Story 1:** Chronological Schedule View (sidebar daily itinerary), Sequential Waypoint Logic for handling multiple classes in a single route, Multi-path Rendering for continuous map paths, and Walk Estimates (time/distance calculations) via the Google Routes API.
- **User Story 2:** Automated Daily Data Synchronization (GitHub Action scheduler running scraper .py every 24 hours) and Backend API Proxy & Security (moving Google Maps keys to environment variables and building a secure Go proxy route).
- **User Story 3:** Calendar (.ics) export support comprising RFC 5545 data schema definitions, Go stream generator for SQLite course records, dynamic UCSC term recurrence logic (ignoring holidays/finals), and the Frontend calendar sync export integration.
- **User Story 4:** Responsive and polished UI with Mobile Navigation Optimization, comprehensive Touch Target & CSS Audit (dark mode spacing and mobile standards), State Hardening (LocalStorage persistence across caches), and custom Sammy the Slug iconography branding.

Work not completed

- **None.** All 14 tasks identified in the Sprint 4 Plan were completed.

Work completion rate

- **Total number of user stories completed (this sprint):** 4
- **Total number of estimated ideal work hours completed (this sprint):** 66 hours
- **Total number of days during this sprint:** 14 days
- **User stories per day (this sprint):** 0.286
- **Ideal work hours per day (this sprint):** 4.71
- **Average user stories per day (across all sprints):** 0.304
- **Average ideal work hours per day (across all sprints):** 4.84

(Note: The final sprint burnup chart for Sprint 4 is available for viewing in the Sprint 4 Plan. Cumulative totals to date reflect 17 user stories and 271 ideal work hours completed over 56 days.)